UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. Authoring tools help developers organise and edit multimedia elements.
2. A URL identifies the address of a web resource.
3. HTML is used to structure content on web pages.
4. RGB is an additive colour model.
5. Sampling converts a continuous sound wave into digital values.
6. Interlacing displays every line of a video frame at the same time.
7. A scanner is an input device.
8. Streaming allows media to begin playing before the whole file is downloaded.
9. A prototype is an early model of a product.
10. Tweening creates intermediate frames between keyframes.
11. Interactivity gives users control over content and flow.
12. Higher resolution always means a smaller file.
13. A plotter is an input device.
14. The amplitude of a sound wave determines its loudness.
15. A higher sampling rate generally produces a more accurate recording.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Keyframe	B. MIDI	C. Bitmap
D. Target audience	E. Multimedia	F. Hypermedia
G. Virtual reality	H. Storyboard	I. Linear multimedia
J. Interactivity	K. Serif	L. Convergence
M. Card-based tool	N. Scanner	O. Sampling

1. image made of pixels
2. simulated environment
3. visual sequence plan
4. converting sound into digital values
5. font with small strokes at character ends
6. authoring tool organised like pages or cards
7. intended users of a product
8. integration of text, graphics, audio, video and animation
9. presentation without user control of sequence
10. standard for musical notes and instructions
11. linked multimedia elements
12. user control over content and flow
13. merging of computing, media and communications
14. input device that digitises documents
15. important frame in an animation

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PART C (30 MARKS)

Choose the correct answer for each question.

1. A browser is used to:
   A. Print documents only
   B. Scan images
   C. Display web pages
   D. Edit audio

2. Resolution refers to:
   A. Number of pixels in an image
   B. File type
   C. Font size
   D. Colour name

3. Which is a benefit of multimedia in education?
   A. It removes all interaction
   B. It uses only text
   C. It replaces all teachers
   D. It combines visuals and sound

4. Hypertext contains:
   A. Only images
   B. No text
   C. Links to other content
   D. Only audio

5. Which is a 3D animation task?
   A. Kerning
   B. Modelling and rendering
   C. Sampling
   D. Bookmarking

6. A licence gives:
   A. Ownership of all works
   B. Permission to use a work
   C. Free hardware
   D. Unlimited bandwidth

7. Path animation moves an object:
   A. Along a predefined route
   B. Randomly
   C. Without motion
   D. Only vertically

8. Cel animation is created by:
   A. Scanning text
   B. Recording sound
   C. Printing images
   D. Drawing each frame by hand

9. RGB is used mainly for:
   A. Printed output
   B. Audio recording
   C. Video editing only
   D. Screen display

10. Streaming allows media to:
    A. Require full download first
    B. Only play offline
    C. Play while downloading
    D. Remove compression

11. Which is NOT a multimedia element?
    A. Text
    B. Compiler
    C. Audio
    D. Animation

12. Kerning adjusts:
    A. Space between individual character pairs
    B. Font colour
    C. Line length
    D. Page size

13. A prototype is:
    A. A font
    B. An early model of a product
    C. A final release
    D. A codec

14. Card-based authoring tools are suited to:
A. Audio editing only
B. Messages with a beginning and end
C. Live video only
D. 3D modelling only

15. Tweening creates:
A. Sound effects
B. Text fonts
C. In-between frames
D. Video codecs

16. A storyboard is:
A. A visual sequence plan
B. A sound file
C. A video codec
D. A colour model

17. Which is an input device?
A. Monitor
B. Scanner
C. Printer
D. Plotter

18. Hypermedia is best described as:
A. Multimedia with hyperlinks
B. A single video file
C. Text-only media
D. A printed manual

19. Interlacing:
A. Draws every line at once
B. Draws alternate lines of a frame
C. Adds sound
D. Removes colour

20. Keyframes are:
A. Colour profiles
B. Important frames in an animation
C. Audio samples
D. Web links

21. Which colour model is subtractive?
A. RGB
B. HSL
C. HSV
D. CMYK

22. Chroma key is used to:
A. Compress audio
B. Add text
C. Scan images
D. Replace a colour background

23. Which format is commonly used for bitmap images?
A. MPEG
B. PNG
C. HTML
D. MIDI

24. A bitmap image is made of:
A. Sound samples
B. Text characters
C. Pixels
D. Mathematical objects

25. MIDI stores:
A. Musical notes and instructions
B. Digitized audio samples
C. Images
D. Video frames

26. RAM is:
A. An input device
B. A web browser
C. Temporary working memory
D. Permanent storage

27. MPEG is:
A. An image editor
B. A storage device
C. A video compression family
D. A font

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28. Frame rate is:

A. Frames per second
B. Bits per second
C. Pixels per inch
D. Samples per second

29. Which is an output device?

A. Plotter
B. Scanner
C. Keyboard
D. Mouse

30. Which is an example of linear multimedia?

A. An interactive kiosk
B. A website with menus
C. A game with choices
D. A cinema presentation

PART D (40 MARKS)

QUESTION 1

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(4 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 2

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 3

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(5 marks)

QUESTION 4

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

END OF QUESTION PAPER